

Diagnostic report -- blind-severe-critical

Overall status: **PAGE** Generated at 2026-08-30T18:04:08.799241+00:00 | Cycle (most recent signal): 2026-08-30T18:04:08.789240+00:00

Executive summary

Data drift: 5/8 feature(s) drifted (severity critical). Target drift (confirmed): PSI=0.000, severity warning. Performance: severity critical, cause = model_weights. Alerts: 3 critical, 1 page, 2 warning. 2 remediation action(s) pending human approval.

Probable root cause identified (cross-agent corroboration)

Probable root cause identified for the performance degradation

Performance degradation (model_weights) coincides with confirmed drift on Glucose, SkinThickness, Insulin, feature(s) with direct SHAP impact: likely causal link.

Recommended action: Prioritize fixing (data or pipeline) Glucose, SkinThickness, Insulin before any blind retrain -- retraining without fixing the cause would reproduce the same problem next cycle.

1. Data / Feature Drift

Overall severity: **critical** -- 5/8 feature(s) drifted (62.5%).

Feature	Type	PSI	Wasserstein (norm)	Severity	Drifted	Quality
Pregnancies	numeric	0.0090	0.0174	none	no	ok
Glucose	numeric	2.0510	1.5349	critical	yes	ok
BloodPressure	numeric	3.7542	0.8923	critical	yes	ok
SkinThickness	numeric	0.0339	6.9195	warning	yes	ok
Insulin	numeric	10.6260	2.3314	critical	yes	29.9% missing
BMI	numeric	1.0948	0.9875	critical	yes	ok
DiabetesPedigreeFunction	numeric	0.0074	0.0273	none	no	ok
Age	numeric	0.0122	0.0200	none	no	ok

2. Target Drift

Status: **confirmed** | n_samples = 768

PSI = 0.0000 | severity = **warning** | drifted = yes Wasserstein distance (normalized) = 0.3008 Reference mean = 0.3490 -> current mean = 0.6497 (reference std = 0.4766 -> current std = 0.4771)

3. Performance Drift

Status: **labeled** | severity: **critical** | remediation_type: model_weights | window = 500 samples y_proba coverage over the window: 100.0%

Performance metrics

Metric	Baseline	Current	Delta	Std score	Degraded	Contributes to severity
accuracy	0.9349	0.5860	-0.3489	-37.32	yes	yes
precision	0.9290	0.5888	-0.3402	-34.20	yes	yes
recall	0.9275	0.5889	-0.3385	-35.02	yes	yes
f1	0.9282	0.5860	-0.3422	-36.87	yes	yes
recall_class_1	0.9030	0.6277	-0.2753	-21.46	yes	yes
auc_roc	0.9829	0.6531	-0.3298	-33.55	yes	yes
auc_pr	0.9701	0.5978	-0.3723	-38.38	yes	yes

Calibration metrics

Metric	Baseline	Current	Delta	Std score	Degraded	Contributes to severity
log_loss	0.1954	0.7391	+0.5436	+51.91	yes	yes
brier_score	0.1041	0.5238	+0.4197	+64.31	yes	yes
ece	0.0591	0.1784	+0.1193	+21.71	yes	yes

Alerts raised this cycle (6)

[PAGE] Recall collapse on critical class (recall_class_1) (source: performance_drift)

recall_class_1 = 0.628 (baseline 0.903): below the absolute floor.

Recommended action (expert): Escalate immediately, outside the standard cycle: possible impact on a sensitive business class. Check whether a rollback is viable, notify the business owner, don't wait for the next scheduled retrain cycle.

Remediation type: escalate

[CRITICAL] High model-impact drift on 3 feature(s) (source: data_drift)

Affected features: Glucose, SkinThickness, Insulin. - Glucose: 'Glucose' is at the 88th percentile of SHAP importance among tracked features (threshold=75th, raw importance=0.5135): the drift directly affects predictions. - SkinThickness: 'SkinThickness' is at the 75th percentile of SHAP importance among tracked features (threshold=75th, raw importance=0.2962): the drift directly affects predictions. - Insulin: 'Insulin' is at the 100th percentile of SHAP importance among tracked features (threshold=75th, raw importance=1.5079): the drift directly affects predictions.

Recommended action (expert): FINAL VERDICT -- pipeline issue confirmed, do NOT retrain yet: Insulin: 29.9% missing values this batch. Fix ingestion for Insulin first (schema, upstream source, mapping of new categories), then re-run this cycle on clean data before deciding whether real drift remains.

Remediation type: data_pipeline_investigation

[CRITICAL] Real model performance degradation (source: performance_drift)

Degraded metrics this window: accuracy, precision, recall, f1, recall_class_1, auc_roc, auc_pr.

Recommended action (expert): Retrain recommended. Check first whether a root_cause alert was raised this cycle: if so, fix that feature/pipeline before retraining, or the same issue will recur.

Remediation type: retrain

[CRITICAL] Probable root cause identified for the performance degradation (source: root_cause)

Performance degradation (model_weights) coincides with confirmed drift on Glucose, SkinThickness, Insulin, feature(s) with direct SHAP impact: likely causal link.

Recommended action (expert): Prioritize fixing (data or pipeline) Glucose, SkinThickness, Insulin before any blind retrain -- retraining without fixing the cause would reproduce the same problem next cycle.

Remediation type: retrain

[WARNING] Drift to monitor on 2 feature(s) (source: data_drift)

Affected features: BloodPressure, BMI. - BloodPressure: 'BloodPressure' has low SHAP importance (percentile 25) and isn't correlated with an influential feature: likely negligible model impact despite the

statistical drift. - BMI: 'BMI' has low SHAP importance (percentile 50) and isn't correlated with an influential feature: likely negligible model impact despite the statistical drift.

Recommended action (expert): No immediate action: risk is indirect (correlated with a key feature) or undetermined (missing SHAP/correlation data). Re-evaluate next cycle.

Remediation type: monitor

[WARNING] Target drift confirmed (true labels) (source: target_drift)

The real target distribution has drifted (PSI=0.000, severity=warning).

Recommended action (expert): Monitor next cycles; no retrain yet, PSI stays below the critical threshold.

Remediation type: monitor

Remediation actions (6)

Type	Urgency (SLA)	Status	Approval required	Automatable	Source alert
data_pipeline_investigation	urgent (within 24h)	executed	no	yes	High model-impact drift on 3 feature(s)
monitor	normal (next working cycle)	executed	no	yes	Drift to monitor on 2 feature(s)
monitor	normal (next working cycle)	executed	no	yes	Target drift confirmed (true labels)
escalate	immediate (act now -- business-critical class impacted, do not wait for the next cycle)	executed	no	yes	Recall collapse on critical class (recall_class_1)

retrain	urgent (within 24h) -- superseded by 'Probable root cause identified for the performance degradation'	skipped	yes	yes	Real model performance degradation
retrain	urgent (within 24h)	pending	yes	yes	Probable root cause identified for the performance degradation

Generated automatically by the drift monitoring pipeline.